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ABSTRACT

Friction from solute–solvent interactions governs processes from molecular diffusion to protein folding and is fundamental for understanding molecular dynamics in liquids. While the fluctuation–dissipation relation determines friction and diffusivity via the velocity autocorrelation function, this exact relation is inconvenient for interfacial systems involving extended surfaces. For interfacial systems, alternative approximate friction formulas based on the force autocorrelation function (FACF) have been introduced. However, these approaches face limitations due to the so-called plateau problem, where the FACF integral decays to zero at long times, complicating friction estimation in particular for small systems. We address these challenges by introducing an exact integral method that is based on the FACF and eliminates the plateau problem, ensuring robust convergence even for small systems. Validated through molecular dynamics simulations of molecular diffusion in SPC/E water, our approach accurately yields diffusivity and friction coefficients and enables decomposing diffusivity contributions into electrostatic and Lennard-Jones forces. Our findings provide a framework for estimating friction from molecular simulations and elucidating the dissipative effects of microscopic forces.

I. INTRODUCTION

The analysis of friction is pivotal in understanding and modeling the dynamics of many-body systems in diverse areas of science. In experimental studies and molecular dynamics (MD) simulations, friction is a key parameter for estimating the long time scales of stochastic processes, such as the diffusion of solutes in complex fluids[1] or conformational transitions in biomolecules and proteins.[2,3] Similarly, friction plays a crucial role in the transport of molecules through biological membranes or in the motion of colloidal particles in suspension.[4,5]

The friction coefficient of an object in a solvent, γ, is defined as the ratio of the constant drag force F that acts on the object with mass m to its stationary velocity v relative to the surrounding solvent according to $\gamma = F/v$. For free diffusion, it can be obtained using the Einstein relation,[6,7] $D = k_B T/\gamma$, where D is the diffusivity obtained from the velocity autocorrelation function (VACF), $C^{vv}(t) = \langle v(0)v(t)\rangle$, via the Green–Kubo relation,[8]

$$D = \int_0^\infty ds\, C^{vv}(s), \tag{1}$$

which follows from the general fluctuation–dissipation relation and, therefore, is an exact expression (see Sec. S1 of the supplementary material). Here, $k_B T$ is the product of the Boltzmann constant and temperature and $\langle\cdots\rangle$ denotes an ensemble average.

A general framework to analyze friction uses the memory friction function $\Gamma(t)$ derived from the generalized Langevin equation (GLE), providing insights into non-Markovian dynamics and enabling robust estimates of friction coefficients in systems with multiple relaxation times. The GLE has been extensively validated and applied to various systems, including solutes in complex fluids and biomolecular processes.[9–23]

However, the interpretation of friction coefficients remains an open question.[24–30] In particular, it is often unclear which inter- and intramolecular forces contribute to friction and to what extent.

The autocorrelation function of the total force $F(t) = m\dot{v}(t)$ (FACF), defined as $C^{FF}(t) = \langle F(0)F(t)\rangle$, is the second derivative of the VACF and thereby connects diffusivity to microscopic forces. Although Eq. (1) is foundational, it cannot be used for objects that are confined in an external potential and thereby do not diffuse freely.[1,14,31] Equation (1) is also inconvenient for solid–liquid interfacial systems since extended surfaces in simulations are typically held fixed in space, and the average motion of bulk liquid phases is difficult to determine with high accuracy.[32,33]

Approximate relations based on the running integral over the FACF between a liquid solvent and an object offer an alternative for estimating friction,[32,34–41] according to which

$$\gamma = \frac{1}{k_B T}\int_0^{\tau_p} ds\, C^{FF}(s), \tag{2}$$

where τ_p is the time scale at which the running integral over the FACF reaches a plateau. This approach has been applied extensively to quantify liquid–surface friction for a wide range of systems, from confined fluids to diffusing particles.[36,42–44] It is useful for capturing frictional effects from equilibrium interfacial simulations and also works when long-range interactions are relevant.[39]

However, the accuracy of friction coefficients obtained from FACF plateau values according to Eq. (2) is limited when dealing with small systems that exhibit a quickly decaying FACF,[9,34,45–50] causing the integral in Eq. (2) to decay toward zero rapidly instead of converging to a meaningful plateau value at intermediate times. This phenomenon, commonly known as the plateau problem, occurs for friction between extended surfaces and liquids as well as for the free diffusion of solutes and limits the distinction of genuine frictional behavior from artifacts introduced by the confined geometry and limited particle number, especially in systems with complex or heterogeneous environments.[9,43,45,51–53]

In contrast, for slowly relaxing systems, friction can be estimated from the FACF integral in Eq. (2) because, for these systems, it exhibits a pronounced intermediate-time plateau for a finite range of values of τ_p,[34,45,54] after which the integral decays to zero. However, even here, the choice of τ_p is subtle, especially for systems for which the FACF has multiple zero-crossings. In the asymptotic limit of an infinite bulk system, the plateau method becomes reliable, resulting in a clear and well-defined plateau in the FACF as $\tau_p \to \infty$.[32,36,52,55,56] These different limitations of the expressions in Eq. (1) and Eq. (2) have driven the search for alternative approaches that work for finite systems without relying on time scale separation or Markovian assumptions.[57,58]

In this work, we approach these challenges by an integral expression for the diffusivity that involves the FACF, is equivalent to Eq. (1), and thereby avoids the plateau problem. Unlike previous techniques, our approach ensures convergence even for small systems. Using MD simulations of SPC/E water, we validate the accuracy of our new formulation and demonstrate its advantage over conventional FACF integration methods for different system sizes and variable object mass. Our method is also applicable for simulations with immobilized particles. As an advantage of our method, we can decompose SPC/E water diffusivity contributions into electrostatic and Lennard-Jones components, offering physical insights into the origin of molecular water diffusion. Our method is general and can be applied to any molecular system.

II. THEORY AND METHODS

For the purpose of illustrating the approximate relationship between Eqs. (1) and (2), we describe the motion of a diffusing object (or more generally of a reaction coordinate) by the GLE.[9] For a one-dimensional reaction coordinate x with effective mass m and velocity $v(t) = \dot{x}(t)$, the GLE in quadratic confinement takes the following form:[9,16,60,61]

$$m\dot{v}(t) = -kx(t) - \int_{-\infty}^{t} ds\, \Gamma(t-s)v(s) + F_R(t), \tag{3}$$

where $\Gamma(t)$ is the friction memory kernel, k is the stiffness of the harmonic potential, and $F_R(t)$ is the random force, following a stationary Gaussian process with zero mean and $\langle F_R(t)F_R(s)\rangle = k_B T\Gamma(|t-s|)$.

In our simulations, we consider systems with $k = 0$, but here, we keep a finite k in order to demonstrate the effect of potential confinement on the FACF. Note that the reaction coordinate $x(t)$ is typically associated with the position of the diffusing object, but it can more generally also refer to the difference coordinate between the object and the embedding solvent, as shown in Sec. S1 of the supplementary material. This becomes relevant for finite-size simulation systems or when the mass of the object is taken to be very large or infinity. The integral over the memory kernel $\Gamma(t)$ over time yields the static friction coefficient of the coordinate γ, i.e., $\gamma = \int_0^\infty dt\, \Gamma(t)$. In this work, we consider isotropic fluids, and we thus average over all directions when computing autocorrelation functions.

Applying the Fourier transformation, $\tilde{x}(\omega) = \int_{-\infty}^{\infty} dt\, e^{-i\omega t}\, x(t)$, on the GLE in Eq. (3), the VACF, $C^{vv}(t) = \langle v(0)v(t)\rangle$, follows in frequency space as[14,16,22,62]

$$\tilde{C}^{vv}(\omega) = \frac{k_B T\omega^2\tilde{\Gamma}(\omega)}{|-k + m\omega^2 - i\omega\tilde{\Gamma}_+(\omega)|^2}, \tag{4}$$

where $\tilde{\Gamma}_+(\omega)$ is the Fourier transformation of the single-sided memory kernel defined as $\Gamma_+(t) \equiv \Gamma(t)$ for $t \geq 0$ and $\Gamma_+(t) \equiv 0$ for $t < 0$, and $\tilde{\Gamma}(\omega) = \tilde{\Gamma}_+(\omega) + \tilde{\Gamma}_+(-\omega)$. The VACF is connected to the total force, $F(t) = m\dot{v}(t)$, autocorrelation function (FACF) via

$$C^{FF}(t) = -m^2\frac{d^2}{dt^2}C^{vv}(t). \tag{5}$$

Accordingly, the FACF in frequency space reads

$$\tilde{C}^{FF}(\omega) = \frac{m^2 k_B T\omega^4\tilde{\Gamma}(\omega)}{|-k + m\omega^2 - i\omega\tilde{\Gamma}_+(\omega)|^2}. \tag{6}$$

Equations (4) and (6) enable determining the friction coefficient γ via autocorrelation functions. According to Eq. (4), for free diffusion ($k \to 0$), the friction memory kernel $\tilde{\Gamma}_+(\omega)$ is related to the Fourier-transformed single-sided VACF $\tilde{C}_+^{vv}(\omega)$ via[14,16,22,67]

$$\tilde{C}_+^{vv}(\omega) = \frac{k_B T}{\tilde{\Gamma}_+(\omega) + i\omega m}. \tag{7}$$

In the zero-frequency limit, we obtain $\tilde{C}_+^{vv}(0) = k_B T/\tilde{\Gamma}_+(0)$, which is equivalent to Eq. (1) with $\tilde{\Gamma}_+(0) = \gamma$. However, for simulations of fixed particles, $\tilde{C}^{vv}(\omega)$ cannot be used to infer the friction. In the limit $m \to \infty$, we obtain from Eq. (6) that $\tilde{C}^{FF}(0) = k_B T\,\tilde{\Gamma}(0)$,[62] being analogous to Eq. (2) for $\tau_p \to \infty$. We thus conclude that Eq. (2) strictly holds for infinite system mass, in which case the limit $\tau_p \to \infty$ can be taken.

In the following, we derive an exact integral expression that allows us to extract the diffusivity and friction coefficient from the FACF also for finite mass. This expression is useful for all situations, where the VACF is not available or where it is difficult to evaluate with sufficient precision.

We define the diffusion integral $\mathscr{D}(\tau)$ as

$$\mathscr{D}(\tau) = \int_0^\tau dt\, C^{vv}(t). \tag{8}$$

Integrating the FACF in Eq. (5) twice and substituting into Eq. (8) yields

$$\mathscr{D}(\tau) = \int_0^\tau dt'' \left[C^{vv}(0) - \frac{1}{m^2}\int_0^{t''} dt' \int_0^{t'} dt\, C^{FF}(t) \right] \tag{9}$$

$$= \int_0^\tau dt \left[C^{vv}(0) - \frac{1}{2m^2}(\tau - t)^2\, C^{FF}(t) \right], \tag{10}$$

as derived in Sec. S1 of the supplementary material. The first term of the integrand in Eq. (10) is the ideal contribution for a non-interacting system, which, according to the equipartition theorem, is constant. As the diffusion integral in Eq. (10) converges in the long-time limit to the diffusivity D in Eq. (1), i.e., $D = \lim_{\tau\to\infty}\mathscr{D}(\tau)$, we immediately see that the second integral in Eq. (10) depending on the FACF also has to scale linearly for long times. Thus, the integral over the second moment of the force acting on a particle cancels the ideal contribution, and Eq. (10) in the long-time limit enables computing the friction coefficient $\gamma = k_B T/D$ solely from the force autocorrelation function and from $C^{vv}(0)$.

The Green–Kubo relation in Eq. (1) and the diffusion integral in Eq. (10) are derived and formulated for the position of a particle x, but they can be equally formulated for the particle–solvent relative coordinate $\Delta x(t)$, which yields a system-size dependent correction term to the expressions in Eqs. (1) and (10) for the diffusivity (see Sec. S1 of the supplementary material for details). According to the equipartition theorem, and considering total-momentum conservation, the mean-squared velocity of the particle with mass m used in Eq. (10) is given by $C^{vv}(0) = k_B T\left(\frac{1}{m} - \left(\frac{1}{m+m_w N}\right)\right)$, where m_w is the mass of the N solvent particles, as derived in Sec. S1 of the supplementary material. However, in simulations, the thermostat enforces $C^{vv}(0) = k_B T/m$,[62] which we use in order to obtain convergence of the diffusion integral also for large particle masses m (Sec. S3 of the supplementary material).

In contrast to Eq. (2), Eq. (10) is exact and, therefore, applicable to finite-mass systems. In contrast to Eq. (1) based on the VACF, Eq. (10) can be split into different contributions in the total force F acting on the particle, e.g., Lennard-Jones or electrostatic contributions, as we discuss below.

III. RESULTS

A. Water diffusivity from force correlations

In this study, we investigate diffusing SPC/E water molecules[63] using MD simulations; for details, see Sec. S2 of the supplementary material. Figure 1(a) depicts the computed VACF (upper panel) and FACF (middle panel) of the center of mass of a single SPC/E water molecule, respectively, from MD simulations with a box size of L = 4.1 nm. The total force acting on a water molecule includes Lennard-Jones and electrostatic components, $F = F_{LJ} + F_{el}$. The FACF (blue, middle panel) agrees perfectly with the data computed from the second derivative of the VACF (orange data, middle panel). The corresponding memory kernel $\Gamma(t)$ in Eq. (3) in the lower panel of Fig. 1(a) is determined from trajectories $x(t)$ via a Volterra extraction scheme explained in Refs. 16 and 59 (blue line in the lower panel); the data are validated by results computed by inverting Eq. (7) (orange line). The water molecule diffuses for L = 4.1 nm with a friction coefficient of $\gamma \approx 1.54 \times 10^{-12}$ kg, here obtained from the mean value in the plateau region of the integral over the memory kernel, i.e., $\gamma = \int_0^\infty dt\,\Gamma(t)$, as shown in Fig. 1(b).

From the FACF in Fig. 1(a), we calculate the diffusion integral of the water molecule $\mathscr{D}(\tau)$ according to Eq. (10). The data in Fig. 1(c) perfectly converge to the diffusivity from the Einstein relation (horizontal dashed lines), $D = k_B T/\gamma$, where γ stems from

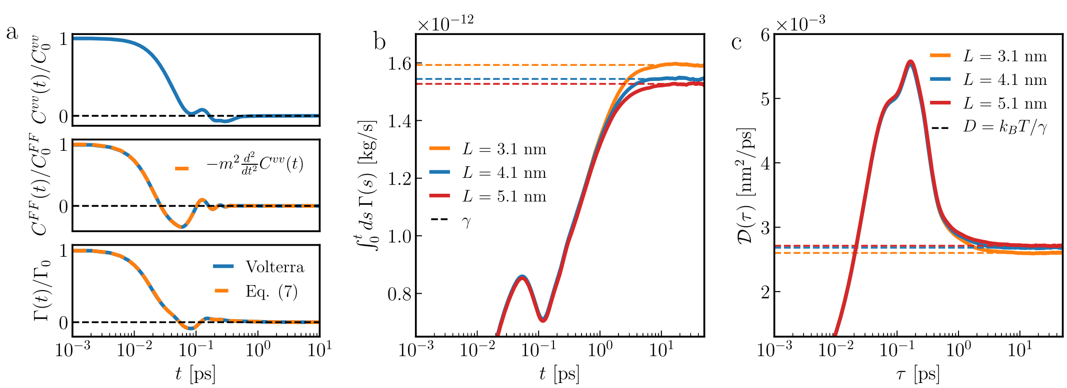

FIG. 1. MD simulation results and diffusivity for a single SPC/E water molecule in liquid water. (a) Extracted velocity autocorrelation function $C^{vv}(t)$ (VACF, upper panel) and force autocorrelation function $C^{FF}(t)$ (FACF, middle panel) for the center of mass of a single SPC/E water molecule from MD simulations (blue data, here for L = 4.1 nm; for details, see Sec. S2 of the supplementary material). The orange line in the middle panel denotes the second numerical derivative of the VACF, demonstrating the accuracy of Eq. (5). The lower panel displays the friction memory kernel $\Gamma(t)$, obtained from Volterra extraction[16,59] (blue) and from the VACF via Eq. (7) (orange). (b) Running integral over the memory kernel obtained from the Volterra scheme for different box sizes L, together with the friction coefficient $\gamma = \int_0^\infty ds\,\Gamma(s)$ estimated by extrapolation (horizontal dashed lines). (c) Diffusion integral $\mathscr{D}(\tau)$ for a water molecule computed from the FACF in [(a), blue] via Eq. (10) for different box sizes L. The horizontal dashed lines denote diffusivities D from the Einstein relation, using γ obtained from the integral over the memory kernels in (b).

the result shown in Fig. 1(b). Thus, the diffusion integral approach to obtain the friction coefficient is numerically robust for molecular diffusion. Our result is not influenced by the thermostat, as we demonstrate in Sec. S4 of the supplementary material. We verify the $L = 4.1$ nm result by comparison with larger and smaller system sizes shown in Figs. 1(b) and 1(c). The diffusivity is known to be influenced by the long-time hydrodynamic tail of the VACF,[64] which can be accounted for by careful consideration of periodic boundary effects.[59] As a result, D increases with increasing system size L.[62,65] However, the finite-size effects for $L = 4.1$ nm are fairly small, as we see only slight deviations between the results for box sizes $L = 4.1$ nm and $L = 5.1$ nm in Fig. 1(c).

In Fig. 2(a), we compare our diffusion integral in Eq. (10) approach with the integration over the FACF according to Eq. (2). Time-integrating the FACF and using Eq. (5), we obtain

$$\int_0^t dt'\, C^{FF}(t') = m^2\left[\dot{C}^{vv}(0) - \frac{d}{dt}C^{vv}(t)\right]. \tag{11}$$

From $\dot{C}^{vv}(0) = 0$ and $C^{vv}(t \to \infty) = 0$, we easily see that as $t \to \infty$, the running integral over the FACF must decay to zero, as confirmed by the orange data in Fig. 2(a). In fact, no plateau is seen for the SPC/E water simulation. This is not surprising since the FACF integral converges only for an infinitely heavy particle embedded in an infinite solvent,[52] as follows from Eq. (6), interpreted in terms of the relative particle–solvent coordinate, in the limit $\omega \to 0$ (see Sec. S5 of the supplementary material for details).

The colored solid lines in Fig. 2(b) depict MD simulation results for a water molecule with variable but finite mass m. As m increases, the decay time of the FACF increases compared to the standard-mass result in black. Nevertheless, the running integrals over the finite-mass FACFs in Fig. 2(c) decay to zero. This underlines that estimating friction coefficients from FACF plateau values according to Eq. (2) is not feasible even for heavy objects. In Sec. S3 of the supplementary material, we further examine the diffusivity for finite-mass systems and demonstrate that Eq. (10) is accurate for calculating the diffusivity, irrespective of the object mass.

For an infinitely heavy particle, i.e., $m \to \infty$, which in simulations is realized by freezing the particle position, the center-of-mass position of the solvent, nevertheless, fluctuates because of total-momentum conservation (Sec. S5 of the supplementary material). The effective mass of the reaction coordinate in this case is given by the total solvent mass, which necessarily is finite in MD simulations. Figure 2(b) shows FACFs for an immobilized water particle ($m \to \infty$) for three different system sizes (dotted lines), which are indistinguishable from the FACF for the system with water mass $m = 10\,000 m_{\mathrm{w}}$. Figure 2(c) demonstrates that the integral over the FACF for infinite water mass $m \to \infty$ decays to zero (dotted lines), which is due to the fact that the effective mass of the reaction coordinate, corresponding to the solvent mean position, is finite. As shown in Sec. S5 of the supplementary material, the method using Eq. (10) is also applicable to systems with a water molecule that is held fixed in space and yields a converging integral. Interestingly, when putting the solvent center-of-mass velocity to zero, the FACF integral in Eq. (2) converges, as demonstrated in Sec. S5 of the supplementary material. There, we explain that this scenario corresponds to applying an infinitely strong harmonic confining potential between particle and solvent.[14]

B. Diffusivity decomposition

The total force acting on the center of mass of a single water molecule consists of electrostatic and Lennard-Jones components, according to $F(t) = F_{\mathrm{el}}(t) + F_{\mathrm{LJ}}(t)$. By inserting these components into Eq. (10) and suitably partitioning the cross correlations, the electrostatic and Lennard-Jones contributions to the diffusivity can be determined, i.e., $D = D_{\mathrm{el}} + D_{\mathrm{LJ}}$.

Figure 3(a) displays the electrostatic (F_{el}) and Lennard-Jones (F_{LJ}) force components from a standard-mass MD simulation with

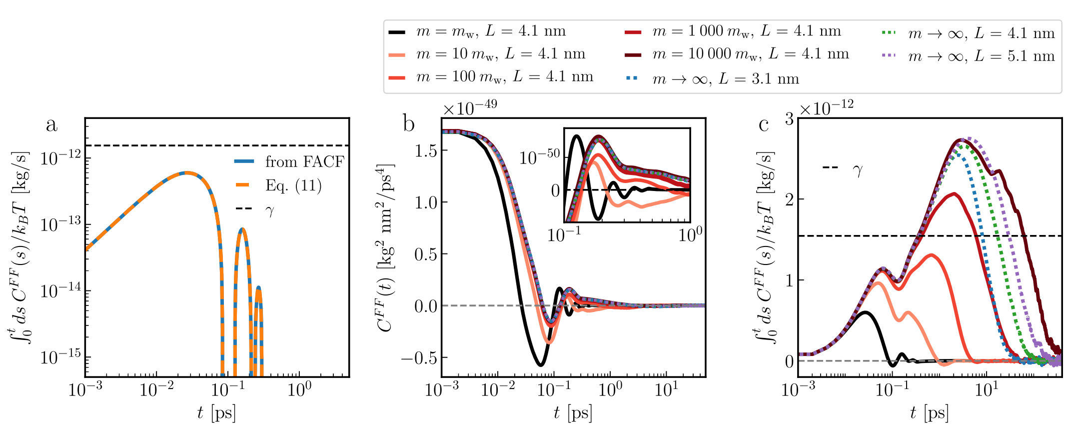

FIG. 2. Friction determined via integration over the FACF. (a) Running integral over the FACF via Eq. (2) (blue), compared with the result using Eq. (11) (orange). The horizontal black dashed line denotes γ computed from the running integral over the memory kernel in Fig. 1(b) for $L = 4.1$ nm. (b) FACFs from MD simulations with a heavy diffusing water molecule ($m = 10 - 10\,000 m_{\mathrm{w}}$, solid lines), here $L = 4.1$ nm, compared with the FACF of a water molecule with standard mass $m_{\mathrm{w}} = 3.01 \times 10^{-26}$ kg (black, $L = 4.1$ nm). The colored dotted lines display the FACF of an immobilized water molecule in SPC/E water (see Sec. S2 of the supplementary material) for different box sizes L. The inset highlights the behavior between 0.1 and 1 ps. Note that in the MD simulations with the fixed molecule, contrary to finite-mass simulations, we do not use center-of-mass velocity removal (see Secs. S2 and S5 of the supplementary material). (c) Running integral over the FACFs shown in (b), divided by k_BT, according to Eq. (2). The horizontal black dashed line denotes the friction coefficient γ computed from the memory kernel in Fig. 1(b).

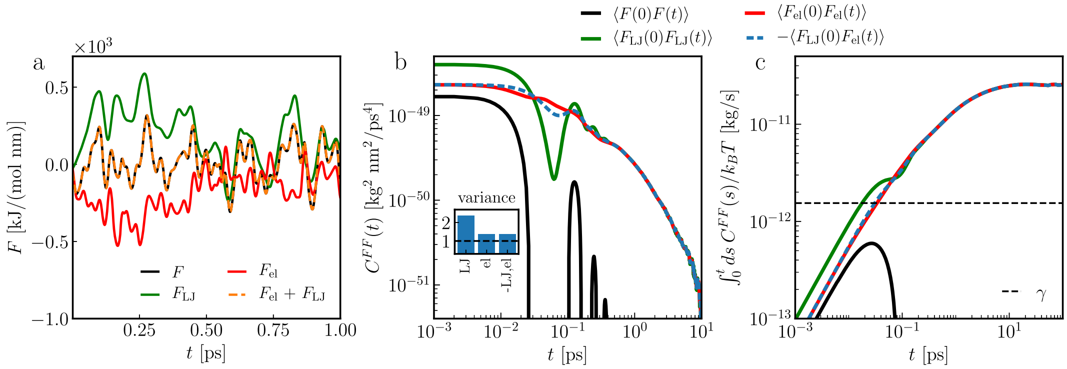

FIG. 3. Decomposition of the total force acting on a single water molecule. (a) Trajectory of the total force acting on an SPC/E water molecule in one Cartesian direction (black), together with the electrostatic (F_{el}, red) and the Lennard-Jones (F_{LJ}, green) components. The sum of both is shown as an orange line. (b) Total FACF [black, same as in Fig. 1(a)], compared with the electrostatic force $\langle F_{el}(0)F_{el}(t)\rangle$ (red) and the Lennard-Jones force $\langle F_{LJ}(0)F_{LJ}(t)\rangle$ (green) autocorrelations, and the cross correlation $\langle F_{LJ}(0)F_{el}(t)\rangle$ (blue dashed line). The inset shows the variances of the forces, normalized by the total force variance $\langle F^2\rangle$. (c) Running integral over the total FACF and the component FACFs shown in (b). The black dashed line denotes the friction coefficient γ determined from the memory kernel $\Gamma(t)$ shown in Fig. 1(b).

$L = 4.1$ nm (see Sec. S2 of the supplementary material). The corresponding FACFs in Fig. 3(b) decay more slowly than for the total force, and the running integrals over these FACFs in Fig. 3(c) do not decay to zero but reach finite long-time values exceeding by far the friction coefficient γ from the memory kernel, taken from Fig. 1(b) and denoted as a horizontal dashed line. Thus, inserting component FACFs into Eq. (10) does not straightforwardly yield plateau values for $\mathscr{D}(\tau)$.

The slower relaxation of the force components can be attributed to the significant negative cross correlation $\langle F_{LJ}(0)F_{el}(t)\rangle$,[26] which matches the magnitude of the autocorrelations, as seen in Fig. 3(b). The force variance follows from $F(t) = F_{LJ}(t) + F_{el}(t)$ as

$$\langle F^2\rangle = \langle F_{LJ}^2\rangle + \langle F_{el}^2\rangle + 2\langle F_{LJ}F_{el}\rangle. \tag{12}$$

For small dipolar molecules, structured solvation layers induce anti-correlation between Lennard-Jones and electrostatic forces.[25,26] For water, we approximately have $\langle F_{LJ}F_{el}\rangle \simeq -\langle F_{el}^2\rangle$,[25,26] confirmed in the inset of Fig. 3(b). Consequently,

$$\langle F^2\rangle \simeq \langle F_{LJ}^2\rangle - \langle F_{el}^2\rangle, \tag{13}$$

which signals an important force balance for water diffusion kinetics.

Accordingly, to ensure the convergence of the diffusion integrals, we split the total FACF into two parts sharing the cross correlation contributions evenly and consider the decomposed FACFs, defined as $C_{el}^{FF}(t) = \langle F_{el}(0)F_{el}(t)\rangle + \langle F_{el}(0)F_{LJ}(t)\rangle$ and $C_{LJ}^{FF}(t) = \langle F_{LJ}(0)F_{LJ}(t)\rangle + \langle F_{LJ}(0)F_{el}(t)\rangle$, which sum up to the total FACF, i.e., $C^{FF}(t) = C_{el}^{FF}(t) + C_{LJ}^{FF}(t)$, and are displayed in Fig. 4(a). Notably, the electrostatic component of the force has zero variance,[26] i.e., $C_{el}^{FF}(0) = 0$. By construction, the running integrals of the component FACFs in Fig. 4(b) vanish for long times.

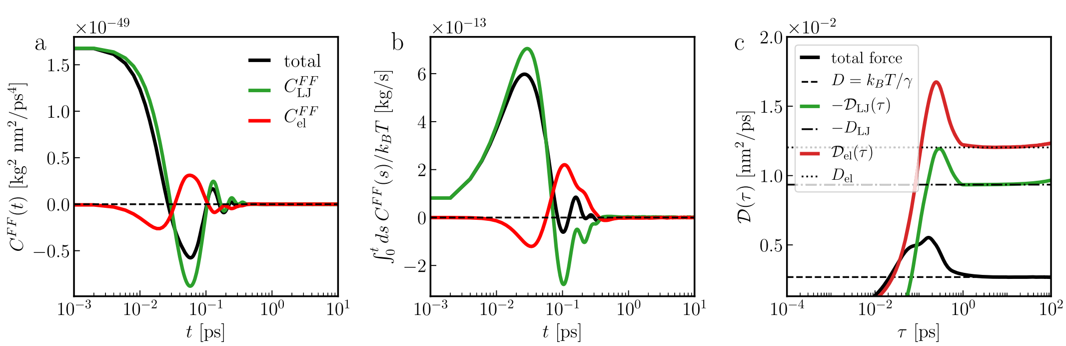

FIG. 4. Decomposition of the FACF and the diffusion integral. (a) Total FACF [black, same as in Fig. 1(a)], compared with the Lennard-Jones FACF $C_{LJ}^{FF}(t)$ (green) and the electrostatic FACF $C_{el}^{FF}(t)$ (red). (b) Running integrals over the total FACF (black) and the component FACFs shown in (a). (c) Diffusion integral in Eq. (10) using the total FACF (black), compared with the diffusivity D from Fig. 1(c) (horizontal dashed line). The green and red lines denote the diffusion integral components from the electrostatic (red) and Lennard-Jones (green) FACFs, obtained by Eq. (14), where $\alpha = 0.239$ is found by the procedure explained in Sec. S6 of the supplementary material. The horizontal dotted and dotted-dashed lines denote the diffusivity components for the electrostatic (D_{el}) and Lennard-Jones force (D_{LJ}), respectively, obtained by the long-time plateaus of the respective diffusion integrals.

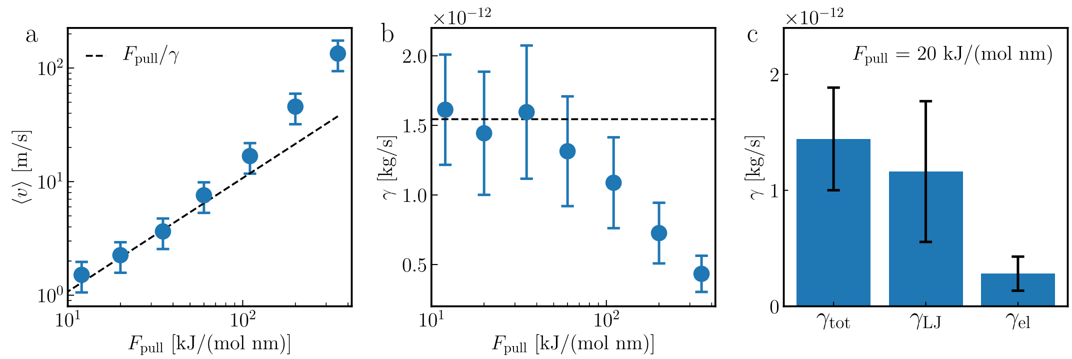

FIG. 5. MD simulation results where a single water molecule is pulled with a constant force $F_{\rm pull}$ in the x-direction (Sec. S7 of the supplementary material) for $L = 4.1$ nm. We show the steady-state velocity $\langle v\rangle$ (a) and the friction coefficient from $\gamma = -\langle F\rangle/\langle v\rangle$ (b) and compare it with the equilibrium result (dashed lines) from Fig. 1(b). Error bars of the velocities and friction coefficients are calculated from ten independent MD simulations with a length of 200 ps each. (c) Decomposition of the friction coefficient, $\gamma_{\rm LJ} = -\langle F_{\rm LJ}\rangle/\langle v\rangle$ and $\gamma_{\rm el} = -\langle F_{\rm el}\rangle/\langle v\rangle$, together with $\gamma_{\rm tot} = \gamma_{\rm LJ} + \gamma_{\rm el}$, here for $F_{\rm pull} = 20$ kJ/(mol nm).

As the diffusion integral in Eq. (10) using the total force for long times equals the diffusivity D, and the ideal contribution gives $\int_0^\tau dt\, C^{vv}(0) = (k_B T/m)\tau$, we define the decomposed diffusion integrals as

$$\begin{aligned}\mathscr{D}_{\rm LJ}(\tau) &= \alpha\frac{k_B T}{m}\tau - \frac{1}{m^2}\int_0^\tau dt''\int_0^{t''}dt'\int_0^{t'}dt\, C^{FF}_{\rm LJ}(t),\\ \mathscr{D}_{\rm el}(\tau) &= (1-\alpha)\frac{k_B T}{m}\tau - \frac{1}{m^2}\int_0^\tau dt''\int_0^{t''}dt'\int_0^{t'}dt\, C^{FF}_{\rm el}(t).\end{aligned} \tag{14}$$

The splitting factor α for the ideal contribution in Eq. (10) is chosen such that the diffusion integrals $\mathscr{D}_{\rm LJ}(\tau)$ and $\mathscr{D}_{\rm el}(\tau)$ converge in the long-time limit. For SPC/E water, we find $\alpha = 0.239$; we give details in Sec. S6 of the supplementary material. Using this α-value, the diffusion integrals $\mathscr{D}_{\rm LJ}(\tau)$ and $\mathscr{D}_{\rm el}(\tau)$ defined in Eq. (14) are shown in Fig. 4(c) and, indeed, are finite in the long-time limit (dotted and dotted-dashed lines). We determine convergence values of $D_{\rm LJ} = -9.35\times10^{-3}$ nm^2/ps and $D_{\rm el} = 12.03\times10^{-3}$ nm^2/ps, both correctly summing up to the total diffusivity $D_{\rm LJ} + D_{\rm el} = 2.68\times10^{-3}$ nm^2/ps obtained by the total diffusion integral in Fig. 1(c). The Lennard-Jones diffusivity contribution $D_{\rm LJ}$ is negative, which is a surprising result that will be contrasted with the decomposition of the friction force acting on a pulled water molecule in Sec. III C. To further illustrate the usefulness of our approach for the decomposition of the diffusivity, we discuss simulations of a Lennard-Jones fluid in Sec. S8 of the supplementary material.

C. Molecular water friction from pulling-force MD simulations

In Fig. 5, we depict results from SPC/E water simulations where we pull one water molecule in a $L = 4.1$ nm box in the x-direction with a constant force $F_{\rm pull}$ (Sec. S2 of the supplementary material). In this non-equilibrium scenario, the average net force acting on the water molecule cancels the pulling force, i.e., $\langle F\rangle = -F_{\rm pull}$, and the molecule moves with the steady-state velocity $\langle v\rangle$ (Sec. S7 of the supplementary material). The steady-state velocity in Fig. 5(a) increases linearly with increasing pulling force for $F_{\rm pull} < 60$ kJ /(mol nm). From this velocity, we compute the friction coefficient via $\gamma = -\langle F\rangle/\langle v\rangle$, as shown in Fig. 5(b). The resulting friction coefficient γ is rather constant for $F_{\rm pull} < 60$ kJ/(mol nm) and agrees well with the equilibrium MD result from Fig. 1(b), shown as a horizontal dashed line.

For $F_{\rm pull} = 20$ kJ/(mol nm), we calculate the mean over the force components $F_{\rm LJ}(t)$ and $F_{\rm el}(t)$ (see Sec. S2 of the supplementary material) and obtain decomposed friction coefficients of $\gamma_{\rm LJ} \equiv -\langle F_{\rm LJ}\rangle/\langle v\rangle \approx (1.16\pm0.6)\times10^{-12}$ kg/s and $\gamma_{\rm el} \equiv \langle -F_{\rm el}\rangle/\langle v\rangle \approx (0.28\pm0.15)\times10^{-12}$ kg/s, both summing up to the total friction coefficient $\gamma_{\rm tot} = (1.44\pm0.44)\times10^{-12}$ kg/s, all visualized in Fig. 5(c). We find that $\gamma_{\rm LJ}$ is much larger than $\gamma_{\rm el}$ and that both friction components $\gamma_{\rm LJ}$ and $\gamma_{\rm el}$ are positive and thus differ qualitatively from the diffusivity components $D_{\rm LJ}$ and $D_{\rm el}$ in Fig. 4(c), where $D_{\rm LJ}$ was found to be negative. This shows that diffusivity and friction components are not individually linked by an analog of the Einstein relation.

IV. CONCLUSION

We provide an exact and robust method to determine the diffusivities and friction coefficients of molecular systems based on force correlations. The method using a weighted integral in Eq. (10) is advantageous over the approach using straight integration of the force autocorrelation in Eq. (2) because it is equivalent to the exact Green–Kubo relation between velocity autocorrelation and diffusivity in Eq. (1), whereas the approximate method via Eq. (2) is only valid in the limit of infinite mass of the reaction coordinate. We verify our approach for simulations with freely diffusing and fixed particles and point out that it can also be applied to interfacial systems.

It is illuminating to contrast the plateau problem, encountered where applying Eq. (2) to finite simulation systems, to the so-called Stokes paradox. The Stokes paradox refers to the finding that when the motion of an infinitely long cylinder or of an infinitely extended plate in an unbounded fluid viscous medium is described by the Stokes equation, the steady-state friction vanishes.[66] The Stokes paradox is not observed for a fluid confined between two infinitely extended plates at finite separation, for which the friction per plate area is finite for plates moving relative to each other (Couette flow) or for a fluid moving relative to the plates (Poiseuille flow). In contrast, the plateau problem of Eq. (2) is also observed for these situations if the FACF is obtained for a finite-mass reaction coordinate.

Our method opens up the possibility of investigating contributions to the diffusivity from different force components, revealing, for SPC/E water, a Lennard-Jones contribution D_{LJ} that is negative. This result should be interpreted with care, since in the FACF decomposition, we split the Lennard-Jones–electrostatic cross correlation to ensure the convergence of the FACF integral in Eq. (10).

The relation between the diffusivity components $D_{\mathrm{LJ}}, D_{\mathrm{el}}$ and the friction components $\gamma_{\mathrm{LJ}}, \gamma_{\mathrm{el}}$ is not trivial: Non-equilibrium simulations in Fig. 5 disclose that Lennard-Jones and electrostatic components positively contribute to friction, $\gamma_{\mathrm{LJ}}, \gamma_{\mathrm{el}} > 0$, meaning that the diffusivity components do not satisfy $D_{\mathrm{LJ}} = k_B T/\gamma_{\mathrm{LJ}}$ and $D_{\mathrm{el}} = k_B T/\gamma_{\mathrm{el}}$, in contrast to the total diffusivity and total friction coefficient, which satisfy the Einstein relation $D = k_B T/(\gamma_{\mathrm{LJ}} + \gamma_{\mathrm{el}})$. We conclude that the diffusivity, reflecting the net random motion of a particle, weights the competing effects of attractive and repulsive components differently than the friction coefficient. Understanding this subtle difference will help clarify the nuanced interplay between microscopic forces and macroscopic transport properties, particularly when the diffusion integral method is applied to more complex systems than water diffusion.

SUPPLEMENTARY MATERIAL

The supplementary material includes supporting simulation details, theory, and results and is attached as a separate file.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Henrik Kiefer: Investigation (equal); Methodology (equal); Validation (equal); Visualization (equal); Writing – original draft (equal). **Benjamin A. Dalton**: Investigation (supporting); Methodology (supporting). **Roland R. Netz**: Conceptualization (equal); Funding acquisition (equal); Methodology (equal); Supervision (equal); Writing – original draft (equal).

DATA AVAILABILITY

Input files of the MD simulations and custom computer codes for the analyses are available from the corresponding author upon reasonable request.

REFERENCES

[1] R. Zwanzig and A. K. Harrison, “Modifications of the Stokes–Einstein formula,” J. Chem. Phys. **83**(11), 5861–5862 (1985).

[2] A. Ansari, C. M. Jones, E. R. Henry, J. Hofrichter, and W. A. Eaton, “The role of solvent viscosity in the dynamics of protein conformational changes,” Science **256**(5065), 1796–1798 (1992).

[3] G. S. Jas, W. A. Eaton, and J. Hofrichter, “Effect of viscosity on the kinetics of α-helix and β-hairpin formation,” J. Phys. Chem. B **105**(1), 261–272 (2001).

[4] P. J. A. M. Kerkhof, “A modified Maxwell-Stefan model for transport through inert membranes: The binary friction model,” Chem. Eng. J. Biochem. Eng. J. **64**(3), 319–343 (1996).

[5] R. R. Nair, H. A. Wu, P. N. Jayaram, I. V. Grigorieva, and A. K. Geim, “Unimpeded permeation of water through helium-leak–tight graphene-based membranes,” Science **335**(6067), 442–444 (2012).

[6] A. Einstein, “Zur Theorie der Brownschen Bewegung,” Ann. Phys. **324**(2), 371–381 (1906).

[7] M. von Smoluchowski, “Zur kinetischen Theorie der Brownschen Molekularbewegung und der Suspensionen,” Ann. Phys. **326**, 756–780 (1906).

[8] R. Zwanzig, *Nonequilibrium Statistical Mechanics* (Oxford University Press, Oxford, NY, 2001).

[9] H. Mori, “Transport, collective motion, and Brownian motion,” Prog. Theor. Phys. **33**(3), 423–455 (1965).

[10] C. Hoheisel, “Memory functions and the calculation of dynamical properties of atomic liquids,” Comput. Phys. Rep. **12**(2), 29–66 (1990).

[11] B. Bagchi and D. W. Oxtoby, “The effect of frequency dependent friction on isomerization dynamics in solution,” J. Chem. Phys. **78**(5), 2735–2741 (1983).

[12] M. Canales and G. Sesé, “Generalized Langevin dynamics simulations of NaCl electrolyte solutions,” J. Chem. Phys. **109**(14), 6004–6011 (1998).

[13] A. Carof, R. Vuilleumier, and B. Rotenberg, “Two algorithms to compute projected correlation functions in molecular dynamics simulations,” J. Chem. Phys. **140**(12), 124103 (2014).

[14] J. O. Daldrop, B. G. Kowalik, and R. R. Netz, “External potential modifies friction of molecular solutes in water,” Phys. Rev. X **7**(4), 041065 (2017).

[15] J. O. Daldrop, J. Kappler, F. N. Brünig, and R. R. Netz, “Butane dihedral angle dynamics in water is dominated by internal friction,” Proc. Natl. Acad. Sci. U. S. A. **115**(20), 5169–5174 (2018).

[16] B. Kowalik, J. O. Daldrop, J. Kappler, J. C. F. Schulz, A. Schlaich, and R. R. Netz, “Memory-kernel extraction for different molecular solutes in solvents of varying viscosity in confinement,” Phys. Rev. E **100**(1), 012126 (2019).

[17] R. Satija and D. E. Makarov, “Generalized Langevin equation as a model for barrier crossing dynamics in biomolecular folding,” J. Phys. Chem. B **123**(4), 802–810 (2019).

[18] C. Ayaz, L. Tepper, F. N. Brünig, J. Kappler, J. O. Daldrop, and R. R. Netz, “Non-Markovian modeling of protein folding,” Proc. Natl. Acad. Sci. U. S. A. **118**(31), e2023856118 (2021).

[19] V. Klippenstein, M. Tripathy, G. Jung, F. Schmid, and N. F. A. van der Vegt, “Introducing memory in coarse-grained molecular simulations,” J. Phys. Chem. B **125**, 4931 (2021).

[20] B. A. Dalton, C. Ayaz, H. Kiefer, A. Klimek, L. Tepper, and R. R. Netz, “Fast protein folding is governed by memory-dependent friction,” Proc. Natl. Acad. Sci. U. S. A. **120**(31), e2220068120 (2023).

[21] B. A. Dalton, H. Kiefer, and R. R. Netz, “The role of memory-dependent friction and solvent viscosity in isomerization kinetics in viscogenic media,” Nat. Commun. **15**(1), 3761 (2024).

[22] J. E. Straub, M. Borkovec, and B. J. Berne, “Calculation of dynamic friction on intramolecular degrees of freedom,” J. Phys. Chem. **91**(19), 4995–4998 (1987).

[23] B. J. Berne, M. E. Tuckerman, J. E. Straub, and A. L. R. Bug, “Dynamic friction on rigid and flexible bonds,” J. Chem. Phys. **93**(7), 5084–5095 (1990).

[24] R. F. Grote and J. T. Hynes, “The stable states picture of chemical reactions. II. Rate constants for condensed and gas phase reaction models,” J. Chem. Phys. **73**(6), 2715–2732 (1980).

[25]T. Samanta and D. V. Matyushov, "Dielectric friction, violation of the Stokes-Einstein-Debye relation, and non-Gaussian transport dynamics of dipolar solutes in water," Phys. Rev. Res. **3**(2), 023025 (2021).
[26]D. V. Matyushov, "War and peace between electrostatic and van der Waals forces regulate translational and rotational diffusion," J. Chem. Phys. **157**(8), 080901 (2022).
[27]K. Zinovjev, P. Guénon, C. A. Ramos-Guzmán, J. J. Ruiz-Pernía, D. Laage, and I. Tuñón, "Activation and friction in enzymatic loop opening and closing dynamics," Nat. Commun. **15**(1), 2490 (2024).
[28]T. Iwashita, Y. Uematsu, M. Terazima, and R. Akiyama, "Decomposition of friction coefficients to analyze hydration effects on a $C_{60}(OH)_n$," J. Chem. Phys. **161**(24), 244907 (2024).
[29]B. A. Dalton and R. R. Netz, "pH modulates friction memory effects in protein folding," Phys. Rev. Lett. **133**(18), 188401 (2024).
[30]H. Kiefer, D. Vitali, B. A. Dalton, L. Scalfi, and R. R. Netz, "Effect of frequency-dependent shear and volume viscosities on molecular friction in liquids," Phys. Rev. E **111**(1), 015104 (2025).
[31]O. F. Lange and H. Grubmüller, "Collective Langevin dynamics of conformational motions in proteins," J. Chem. Phys. **124**(21), 214903 (2006).
[32]J. Petravic, "Force autocorrelation function in linear response theory and the origin of friction," J. Chem. Phys. **129**(9), 094503 (2008).
[33]A. T. Bui and S. J. Cox, "Revisiting the Green–Kubo relation for friction in nanofluidics," J. Chem. Phys. **161**(20), 201102 (2024).
[34]J. G. Kirkwood, "The statistical mechanical theory of transport processes I. General theory," J. Chem. Phys. **14**(3), 180–201 (1946).
[35]R. M. Mazo, "On the theory of Brownian motion. III. Two-body distribution function," J. Stat. Phys. **1**(4), 559–562 (1969).
[36]P. Mazur and I. Oppenheim, "Molecular theory of Brownian motion," Physica **50**(2), 241–258 (1970).
[37]J. M. Deutch and I. Oppenheim, "Molecular theory of Brownian motion for several particles," J. Chem. Phys. **54**(8), 3547–3555 (1971).
[38]T. J. Murphy and J. L. Aguirre, "Brownian motion of *N* interacting particles. I. Extension of the Einstein diffusion relation to the *N*-particle case," J. Chem. Phys. **57**(5), 2098–2104 (1972).
[39]L. Bocquet and J.-L. Barrat, "Flow boundary conditions from nano- to micro-scales," Soft Matter **3**(6), 685–693 (2007).
[40]K. Huang and I. Szlufarska, "Green–Kubo relation for friction at liquid–solid interfaces," Phys. Rev. E **89**(3), 032119 (2014).
[41]J. A. de la Torre, D. Duque-Zumajo, D. Camargo, and P. Español, "Microscopic slip boundary conditions in unsteady fluid flows," Phys. Rev. Lett. **123**(26), 264501 (2019).
[42]J.-P. Hansen and I. R. McDonald, *Theory of Simple Liquids: With Applications to Soft Matter* (Academic Press, Amsterdam, Boston, 2013).
[43]L. Bocquet, J.-P. Hansen, and J. Piasecki, "On the Brownian motion of a massive sphere suspended in a hard-sphere fluid. II. Molecular dynamics estimates of the friction coefficient," J. Stat. Phys. **76**, 527–548 (1994).
[44]K. Falk, F. Sedlmeier, L. Joly, R. R. Netz, and L. Bocquet, "Molecular origin of fast water transport in carbon nanotube membranes: Superlubricity versus curvature dependent friction," Nano Lett. **10**(10), 4067–4073 (2010).
[45]A. N. Lagar'kov and V. M. Sergeev, "Molecular dynamics method in statistical physics," Sov. Phys.-Usp. **21**(7), 566 (1978).
[46]F. Ould-Kaddour and D. Levesque, "Determination of the friction coefficient of a Brownian particle by molecular-dynamics simulation," J. Chem. Phys. **118**(17), 7888–7891 (2003).
[47]L. Bocquet and J.-L. Barrat, "On the Green–Kubo relationship for the liquid–solid friction coefficient," J. Chem. Phys. **139**(4), 044704 (2013).
[48]S. Chen, H. Wang, T. Qian, and P. Sheng, "Determining hydrodynamic boundary conditions from equilibrium fluctuations," Phys. Rev. E **92**(4), 043007 (2015).
[49]H. Oga, Y. Yamaguchi, T. Omori, S. Merabia, and L. Joly, "Green–Kubo measurement of liquid–solid friction in finite-size systems," J. Chem. Phys. **151**(5), 054502 (2019).
[50]H. Oga, T. Omori, C. Herrero, S. Merabia, L. Joly, and Y. Yamaguchi, "Theoretical framework for the atomistic modeling of frequency-dependent liquid–solid friction," Phys. Rev. Res. **3**(3), L032019 (2021).
[51]A. Suddaby and P. Gray, "Relations between the friction constant and the force correlation integral in Brownian movement theory a comparison of the microscopic and macroscopic theories," Proc. Phys. Soc. **75**(1), 109 (1960).
[52]P. Español and I. Zúñiga, "Force autocorrelation function in Brownian motion theory," J. Chem. Phys. **98**(1), 574–580 (1993).
[53]H. Oga, T. Omori, L. Joly, and Y. Yamaguchi, "Equilibrium molecular dynamics evaluation of the solid–liquid friction coefficient: Role of timescales," J. Chem. Phys. **159**(2), 024701 (2023).
[54]R. Vogelsang and C. Hoheisel, "Determination of the friction coefficient via the force autocorrelation function. A molecular dynamics investigation for a dense Lennard-Jones fluid," J. Stat. Phys. **47**, 193–207 (1987).
[55]J. L. Lebowitz and E. Rubin, "Dynamical study of Brownian motion," Phys. Rev. **131**(6), 2381 (1963).
[56]P. Resibois and H. T. Davis, "Transport equation of a Brownian particle in an external field," Physica **30**(6), 1077–1091 (1964).
[57]P. Español, J. A. de la Torre, and D. Duque-Zumajo, "Solution to the plateau problem in the Green–Kubo formula," Phys. Rev. E **99**(2), 022126 (2019).
[58]H. Nakano and S.-i. Sasa, "Microscopic determination of macroscopic boundary conditions in Newtonian liquids," Phys. Rev. E **99**(1), 013106 (2019).
[59]L. Scalfi, D. Vitali, H. Kiefer, and R. R. Netz, "Frequency-dependent hydrodynamic finite size correction in molecular simulations reveals the long-time hydrodynamic tail," J. Chem. Phys. **158**(19), 191101 (2023).
[60]R. Zwanzig, "Memory effects in irreversible thermodynamics," Phys. Rev. **124**(4), 983 (1961).
[61]C. Ayaz, L. Scalfi, B. A. Dalton, and R. R. Netz, "Generalized Langevin equation with a nonlinear potential of mean force and nonlinear memory friction from a hybrid projection scheme," Phys. Rev. E **105**, 054138 (2022).
[62]J. O. Daldrop and R. R. Netz, "Mass-dependent solvent friction of a hydrophobic molecule," J. Phys. Chem. B **123**(38), 8123–8130 (2019).
[63]H. J. C. Berendsen, J. R. Grigera, and T. P. Straatsma, "The missing term in effective pair potentials," J. Phys. Chem. **91**(24), 6269–6271 (1987).
[64]M. H. Ernst, E. H. Hauge, and J. M. J. van Leeuwen, "Asymptotic time behavior of correlation functions," Phys. Rev. Lett. **25**(18), 1254 (1970).
[65]I.-C. Yeh and G. Hummer, "System-size dependence of diffusion coefficients and viscosities from molecular dynamics simulations with periodic boundary conditions," J. Phys. Chem. B **108**(40), 15873–15879 (2004).
[66]H. Lamb, *Hydrodynamics* (Cambdrige University Press, 1932).
[67]B. J. Berne, J. P. Boon, and S. A. Rice, "On the calculation of autocorrelation functions of dynamical variables," J. Chem. Phys. **45**(4), 1086–1096 (1966).